وزارة الاتصالات وتقنية المعلومات MINISTRY OF COMMUNICATIONS AND INFORMATION TECHNOLOGY | الأكاديمية السعودية الرقمية SAUDI DIGITAL ACADEMY | صندوق تنمية الموارد البشرية HUMAN RESOURCES DEVELOPMENT FUND | المؤسسة العامة للتدريب التقني والمهني Technical and Vocational Training Corporation | مهارات المستقبل | معسكرات همة الرقمية 2025 | Clarusway

شهادة إتمام تدريب

CERTIFICATE OF COMPLETION

تبارك الأكاديمية السعودية الرقمية لـ

Norah Khaled Alshdoukhi

نورة خالد الشدوخي

On successfully completing **Software Testing Bootcamp** in a journey that lasted **13 weeks**
From **24 August 2025** to **20 November 2025**

إتمام معسكر **همة لاختبار البرمجيات** بنجاح بعد رحلة استمرت لمدة **13 أسبوع** في الفترة من **1 ربيع الأول 1447** الى **29 جمادى الأول 1447**

Issue date **20 November 2025**

تم إصدارها بتاريخ **29 جمادى الأول 1447**

WISHING YOU A CONTINOUS SUCCESS

مع تمنياتنا بدوام التوفيق والنجاح

Amal A. Aljuhaiman

الرئيس التنفيذي للأكاديمية السعودية الرقمية

امل عبدالكريم الجهيمان



Bootcamp Syllabus

WEEK 1

Database

- Introducing database concepts
- Database management system
- Relations
- Tables-SQL-no SQL
- Commands
- Joins

WEEK 2

Automation

- Java Database Connectivity (JDBC)
- Software Testing Life Cycle (STLC)
- Selenium
- JUnit

WEEK 3

Front-end testing

- XML Path Language (XPath)
- (WebElement) Commands
- Locators Practice
- ChromeOptions
- Iframe
- SendKeys & Clear
- Selenium Waits

WEEK 4

Web application Testing

- TestBase
- WindowHandle
- Actions
- JavaScriptExecutor
- File handling
- Excel handling
- Robot Class
- Cookies
- ScreenShot
- ExtentReports

WEEK 5

TestNG Framework

- TestNG Framework
- TestNG Annotations
- TestNG Assertions
- XML Files
- DataProvider
- CrossBrowser
- Parallel Testing
- Smoke Test
- Allure Reports

WEEK 6

TestNG Framework

- Listeners
- DriverPageClass
- POM Practice
- Singleton & Parallel Test

WEEK 7

Testing Framework

- Cucumber Framework & Gherkin Language
- RunnersClass & ScenarioOutline
- Hooks & Failed Scenario & Data Table
- Runner Class & Parameterized Feature
- Fluent POM

WEEK 8

Testing Framework

- Selenium Grid
- Jenkins:
  - Introduction to EC2
  - Jenkins Pipeline
  - JenkinsFile_Maven Project

Bootcamp Syllabus

WEEK 9

API testing

- API
- What's HTTP's-Swaggwer-CRUD
- Authorization
- Postman
- RestAssured

WEEK 10

API testing & Implementing AI

- aseURI – PathParams – QueryParams – Validation
- JSON Object
- ObjectMapper-Serialization-Deserialization
- POJO Class
- Create an API Test Automation Framework from Scratch
- Introduction to AI
- Implementing AI for Automated Testing
- AI-based Test Analysis and Maintenance
- AI Applications in Testing
- AI and the Future of SDET
- Students apply everything learned in a final project, simulating real-worldtest scenarios.
- Jira Overview
- Bug Reporting
- Test Execution in Jira
- Allure Reports
- Project Retrospective

WEEK 11

UI

- Management Software for Capstone Project
- Real Business Environment
- Role of Tester & Other Team Members
- Software Development Life Cycle (SDLC)
- Software Testing Life Cycle (STLC)
- Agile Methodologies
- Scrum Events
- Planning and Management of Test Automation
- User Stories & Acceptance Criteria's
- Cucumber Behavior Driven Development (BDD) framework
- Different Test Types (Smoke, Happy Path, End-to-end testing)
- Various Scenarios for Testing (positive - Negative)
- Bug life cycle
- Project version control system by using GitHub
- Jenkins Using
- Manual Testing

WEEK 12

API

- Management Software for Capstone Project
- Real Business Environment
- Role of Tester & Other Team Members
- Software Development Life Cycle (SDLC)
- Software Testing Life Cycle (STLC)
- Agile Methodologies
- Scrum Events
- Planning and Management of Test Automation

WEEK 13

API

- User Stories & Acceptance Criteria's
- Cucumber Behavior Driven Development (BDD) framework
- Different Test Types (Smoke, Happy Path, End-to-end testing)
- Various Scenarios for Testing (positive - Negative)
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